Strategic Decision Report: Data-Driven Selection of an Offshore Clinical Data Delivery Center for XYZ Clinical Data Solutions

1. Introduction, About the Company, and Scenario Description

About XYZ Clinical Data Solutions: XYZ Clinical Data Solutions is a United States-based provider of clinical data management services to small and mid-sized pharmaceutical, biotechnology, and medical device companies conducting Phase I–IV clinical trials. Its services include building the electronic data capture (EDC) systems used to collect trial data, cleaning and validating that data, medical coding, and preparing standardized datasets for regulatory submission. The company is also expanding a line of AI-assisted services, including automated detection of data discrepancies. XYZ currently employs approximately 350 staff and supports around 40 active sponsor studies.

The Expansion Scenario: XYZ is evaluating its first international expansion: an offshore clinical data delivery center of 50–75 employees. The center would initially handle data cleaning, query management, and medical coding, while client-facing project management and regulatory submission support would remain in the United States. It would not serve local customers. Its value proposition is to expand delivery capacity, support faster turnaround by extending data-cleaning activities across time zones, and provide access to skilled data professionals who are increasingly scarce and costly in the United States, while maintaining the data quality and security that sponsors expect. Because XYZ manages sensitive and regulated patient data, the location decision depends not only on talent availability and cost, but also on data security practices, business digital infrastructure, and openness to foreign investment and cross-border digital services. This report presents the completed analysis supporting that decision.

Summary of Analysis and Recommendations: Forty-three candidate countries were evaluated against nine OECD Going Digital indicators organized into four strategic pillars. Thirty-one countries had sufficient data to receive an overall readiness score; sixteen carry at least one risk signal. Three findings drive the recommendations.

First, Finland (89.5), Sweden (85.7), and Ireland (84.8) emerge as the most robust candidates. Finland and Sweden remain in the top five under every weighting scenario tested, and Ireland does so under three of four. None of the three carries a risk flag.

Second, Korea ranks fourth overall (85.5) but is classified by the OECD among the three lowest-performing countries on digitally-deliverable services exports, the single indicator most directly tied to cross-border service delivery. Germany ranks third under baseline weights but first when talent is prioritized, making its position dependent on management's stated priorities rather than robust to them.

Third, the countries that can be fully scored are concentrated in high-income economies. Several emerging markets that would ordinarily feature in an offshore delivery strategy, including India, Brazil, Colombia, and Costa Rica, lack sufficient OECD coverage to be ranked at all. Cost competitiveness is not visible anywhere in this dataset and must be established separately.

Accordingly, this report recommends that XYZ (1) advance Finland, Sweden, and Ireland to full due diligence; (2) retain Korea and Estonia as conditional candidates pending targeted validation; and (3) commission supplementary cost, language, and clinical-research data before excluding any data-sparse market.

2. Objective of the Decision-Making Project

Purpose: The purpose of this project is to identify, on evidence rather than intuition, which countries are best suited to host XYZ's first offshore clinical data delivery center, and to produce a shortlist that leadership can take forward to detailed cost, regulatory, and operational due diligence.

International expansion exposes firms to significant risk when decisions are made without adequate preparation and evidence. For XYZ the exposure is amplified: a delivery center that cannot recruit and retain qualified staff, meet sponsor quality expectations, or protect regulated patient data would damage client relationships, delay trial timelines, and create regulatory and reputational risk across the entire business, not only at the new site. Because the center is intended to strengthen service to existing sponsors rather than generate new revenue, it must deliver consistent quality from the outset.

Goals of the Analysis: The analysis was designed to achieve four specific goals. The first was to establish which country characteristics best indicate readiness for a regulated, digitally intensive service operation, and to select indicators accordingly rather than adopting general measures of digital development. The second was to compare all candidate countries on a consistent, reproducible basis, so that any recommendation can be traced back to the evidence that produced it. The third was to distinguish countries that perform poorly from countries about which little is known, so that promising markets are not eliminated by gaps in a dataset. The fourth was to test whether the resulting shortlist is stable when management priorities change, rather than an artifact of one particular set of assumptions.

Key Decisions Structuring the Expansion: The project was built to resolve three decisions that together structure how XYZ approaches the expansion, each carried forward from the approved proposal and addressed by the completed analysis.

Decision 1: Which indicators define readiness for a regulated offshore delivery center? Candidate indicators were assessed for relevance to XYZ's operating model, country coverage, recency, and overlap, and nine were organized into four strategic pillars. Three relevant but thinly covered measures were retained as supporting evidence rather than scored. The resulting framework is described in the Methodology section and listed in full in Appendix A.

Decision 2: How should countries be compared, grouped, and ranked? Countries are ranked on a weighted composite of the four pillars, grouped by strategic profile through clustering, and tested under alternative weightings to establish which country positions are robust. Countries that cannot be scored are assessed separately rather than ranked alongside those that can. The Methodology section describes how each element was built, and the Results section reports what it produced.

Decision 3: Which conditions should trigger caution or remove a country from consideration? A two-tier screen distinguishes observed weakness from missing information. Risk flags draw on OECD bottom rankings and bottom-quartile pillar performance, while countries lacking a composite score are identified as evidence gaps requiring further inquiry rather than penalized. The Methodology section sets out the thresholds, and the Results section identifies the countries affected.

Assumptions and Constraints: Several assumptions framed the analysis. The United States was excluded as XYZ's home market. The entry model assumes a wholly owned delivery center rather than outsourcing to a third-party provider, although a partnership may be reconsidered where foreign-investment restrictions prove material. Because the center will not serve local customers, local market size and saturation are not relevant to the decision. An indicative investment envelope of approximately US$4–6 million over the first two years, and a phased ramp to 50–75 staff, are treated as company assumptions.

Two constraints shape how the results should be read. The OECD dataset measures digital readiness but does not capture labor costs, English-language proficiency, time-zone compatibility, political stability, or the depth of the local clinical-research workforce. And indicator coverage is uneven: the dataset is substantially more complete for high-income European economies than for emerging markets, which systematically limits which countries can be ranked. Both constraints are addressed explicitly in the methodology rather than left implicit.

The Value of a Data-Driven Approach: A structured analytical approach delivers three things that judgment alone cannot. It replaces general impressions about popular outsourcing destinations with comparable

evidence across every candidate country. It makes XYZ's priorities explicit, by tying each pillar to a specific operating requirement, so that countries are judged against what the center actually needs rather than against general digital performance. And it exposes the reasoning to challenge: because the weighting is adjustable and the underlying indicators are visible, leadership can test the recommendation against their own assumptions rather than accept or reject it as a single number.

3. Methodology

Data Source and Preparation: The analysis uses the OECD Going Digital Toolkit dataset (OECD, 2025), which reports standardized digital-economy indicators across 47 entries. The primary data preparation and analysis were carried out in Tableau, with selected results independently validated outside it.

Three aggregate rows (EU27, G20, and the OECD average) were excluded because they are not candidate locations, and the United States was excluded as the home market, leaving 43 candidate countries.

Each indicator appears twice in the dataset: as a raw Value in its native unit, and as a Distance score scaled from 0 to 100 against the best-performing country. The analysis uses Distance scores so that indicators expressed in different units, such as percentages, index values, and shares of GDP, can be combined on a common scale. A direction check confirmed that higher Distance scores consistently represent more favorable performance, including for inversely-scaled indicators: Canada records the lowest rate of privacy violations and a Distance score of 100, while Chile records a higher violation rate and a score of 93. The same relationship holds for the FDI and Digital Services Trade Restrictiveness indices, where the least restrictive country scores highest.

Data Cleaning: Duplicate Records: An audit of record counts identified 42 duplicated country–indicator pairs. These arise because the dataset assigns more than one classification to the same observation: Korea, for example, is recorded as both “Bottom 2” and “Fastest changing 2” on digitally-deliverable services exports, producing two identical rows.

Left uncorrected, these records distort any aggregation. A deduplication flag was created to retain a single record per country–indicator pair, and applied as a context filter so that it executes before Tableau's level-of-detail calculations rather than after them. The correction was material rather than cosmetic: Korea's Market Openness score rose from 60.97 to 70.98, moving the country from sixth to fourth place in the overall ranking.

Scoring Framework: Nine core indicators were organized into four strategic pillars using Tableau groups (Appendix A). Pillar scores were built as calculated fields using level-of-detail expressions that average the available Distance scores for each country within each pillar, returning a null value where a country has no data in that pillar.

An overall Composite Readiness score combines the four pillars using weights controlled by Tableau parameters, with a baseline of Talent and Workforce 35 percent, Trust and Data Security 25 percent, Business Digital Infrastructure 20 percent, and Market Openness 20 percent. These weights reflect XYZ's operating priorities rather than any property of the data, which is precisely why they are exposed as adjustable controls rather than embedded as fixed constants. The calculation normalizes by the sum of the weights, so rankings remain valid at any slider position.

A country is treated as fully scorable when at least one core indicator is available in every pillar. Thirty-one countries meet this test. The remaining twelve are retained as data-sparse candidates and assessed separately on the pillars for which data exists, so that absent evidence is not mistaken for weak performance.

Risk Screening: Two risk measures were constructed. The first draws on the OECD's own “Bottom 1–3” classifications for core Trust and Market Openness indicators, providing a source-based rather than arbitrary

threshold. The second flags countries falling within the bottom quartile of scorable countries on Trust (at or below 77.8), Market Openness (70.8), or Infrastructure (66.6).

Countries without a composite score are classified separately as “insufficient data” rather than flagged as risky. This distinction proved consequential. An earlier version that did not separate them returned a risk count of 25, because data-sparse countries such as Japan and India were being penalized for missing indicators rather than for observed weakness. The corrected measure identifies 16 countries carrying genuine risk signals.

Strategic Segmentation: Tableau's k-means clustering was applied to the four pillar scores across the 31 fully scorable countries, producing three groups. The clusters describe strategic profiles rather than a ranking: high readiness across all four pillars (13 countries), balanced mid-tier performance (15 countries), and an infrastructure-limited profile (3 countries) combining strong trust scores with very weak business infrastructure.

Validation: Five validation steps were performed. Cluster assignments were reproduced using an independent k-means calculation outside Tableau, and all 31 country assignments matched exactly. Country names were validated against Tableau's geographic roles, where one ambiguous entry (“Korea”) required manual mapping to the Republic of Korea. Aggregation behavior was verified after an early sort produced incorrect ordering, which traced to summing rather than averaging pillar scores in a view containing multiple rows per country. The clustering and the risk screen were compared as independent methods: none of the 13 high-readiness countries carries a below-quartile flag, which supports the internal consistency of both. Finally, because a country can qualify for a composite score with as few as one indicator per pillar, the baseline ranking was re-run using only the 24 countries with complete data on all nine indicators. Finland, Sweden, Germany, and Ireland remain the top four in the same order, confirming that the shortlisted countries remain among the strongest candidates when the analysis is restricted to complete cases.

Dashboard Design: An executive dashboard brings together a readiness map, a cluster scatter plot with quadrant reference lines, and a ranked bar chart colored by risk flag, alongside three calculated summary measures and the four weight sliders. Selecting a country filters the other views, so a position in the ranking can be traced immediately to its geographic context and strategic profile. Tableau made the comparison tractable in three specific ways: it held nine indicators of differing units on a single scale, it exposed the weighting assumption as a control rather than burying it in a formula, and it displayed data completeness beside every score so that confidence and performance are read together. The sliders allow management to test the robustness of any shortlist directly, rather than relying on a single fixed score. This is the capability identified in the project proposal as central to evidence-based decision-making. The dashboard is shown at Figure 2.

Supplementary Data: The OECD dataset does not capture several factors material to an offshore-services decision. Four supplementary factors were identified for the shortlisting stage. Two were incorporated into the present analysis, while labor-cost and clinical-research data remains pending acquisition:

Factor	Source	Status
Labor cost, data and technical roles	ILOSTAT (International Labour Organization, n.d.) and national wage statistics	Pending acquisition
English-language proficiency	EF English Proficiency Index	External data
Time-zone overlap with U.S. business hours	Derived from standard UTC offsets	Calculated
Clinical-research activity	ClinicalTrials.gov registered trial counts (National Library of Medicine, n.d.)	Pending acquisition

Table 1. Supplementary data applied at the shortlisting stage.

These factors are applied after the readiness comparison rather than within it, so that the OECD-based scoring remains methodologically clean and the effect of each additional consideration stays visible.

4. Results and Recommendations

The analysis is summarized across two views. The composite readiness ranking is presented first, followed by the executive dashboard at Figure 2, which combines the readiness map, the strategic profiles segmentation, and the ranked bar chart with risk flags, together with the adjustable pillar weights used for sensitivity testing. Detailed views of each component appear in Appendix C.

Composite Readiness Ranking: Under baseline weights, the twelve highest-scoring countries are shown below. Data completeness is reported alongside each score, because a high score supported by partial evidence warrants more caution than the same score supported by complete evidence. The full ranking of all 43 candidate countries appears at Figure C1.

Rank	*Country*	*Composite Readiness*	*Trust Score*	*Talent Score*	*Infrastructure Score*	*Market Openness Score*	*Data Completeness %*
1	Finland	89.5	99.6	81.6	96.8	83.7	100%
2	Sweden	85.7	92.4	78.9	91.5	83.6	100%
3	Germany	85.6	82.6	96.5	76.2	79.8	100%
4	Korea	85.5	95.7	84.8	88.8	71.0	78%
5	Ireland	84.8	88.4	79.2	82.5	92.2	100%
6	Switzerland	83.6	98.2	79.3	71.0	85.3	89%
7	Netherlands	83.2	85.8	76.3	91.5	83.9	100%
8	Denmark	83.2	95.2	74.3	94.4	72.7	100%
9	Canada	83.1	100.0	80.7	69.2	80.2	78%
10	Luxembourg	81.2	83.6	75.1	71.5	98.3	100%
11	Estonia	80.9	80.9	81.4	80.6	80.3	100%
12	United Kingdom	80.0	85.6	78.0	71.3	85.3	100%

Figure 1. Composite readiness ranking under baseline weights. Top twelve countries at Talent 35%, Trust 25%, Infrastructure 20%, Market Openness 20%, with pillar scores and data completeness.

The scores cluster tightly: second through ninth place span fewer than three points. This narrow spread is itself a finding. It means the ranking alone cannot responsibly select a location, and that segmentation, risk screening, and sensitivity testing carry more decision weight than the ordinal position of any single country.

Strategic Segmentation: Clustering on the four pillar scores produced three distinct strategic profiles (Figure 2; the scatter is reproduced at Figure C3 and full membership at Figure C4).

Profile	**n**	**Talent**	**Trust**	**Infra.**	**Open.**	**Interpretation**
High readiness	13	79.5	90.5	82.7	82.6	Strong performance across all four pillars
Balanced mid-tier	15	76.7	76.3	68.7	72.1	Adequate talent; weaker infrastructure
Infrastructure-limited	3	64.5	97.6	38.9	69.3	High trust, infrastructure below half of leaders

Table 2. Cluster profiles with mean pillar scores. Cluster assignments were independently reproduced outside Tableau with identical results.

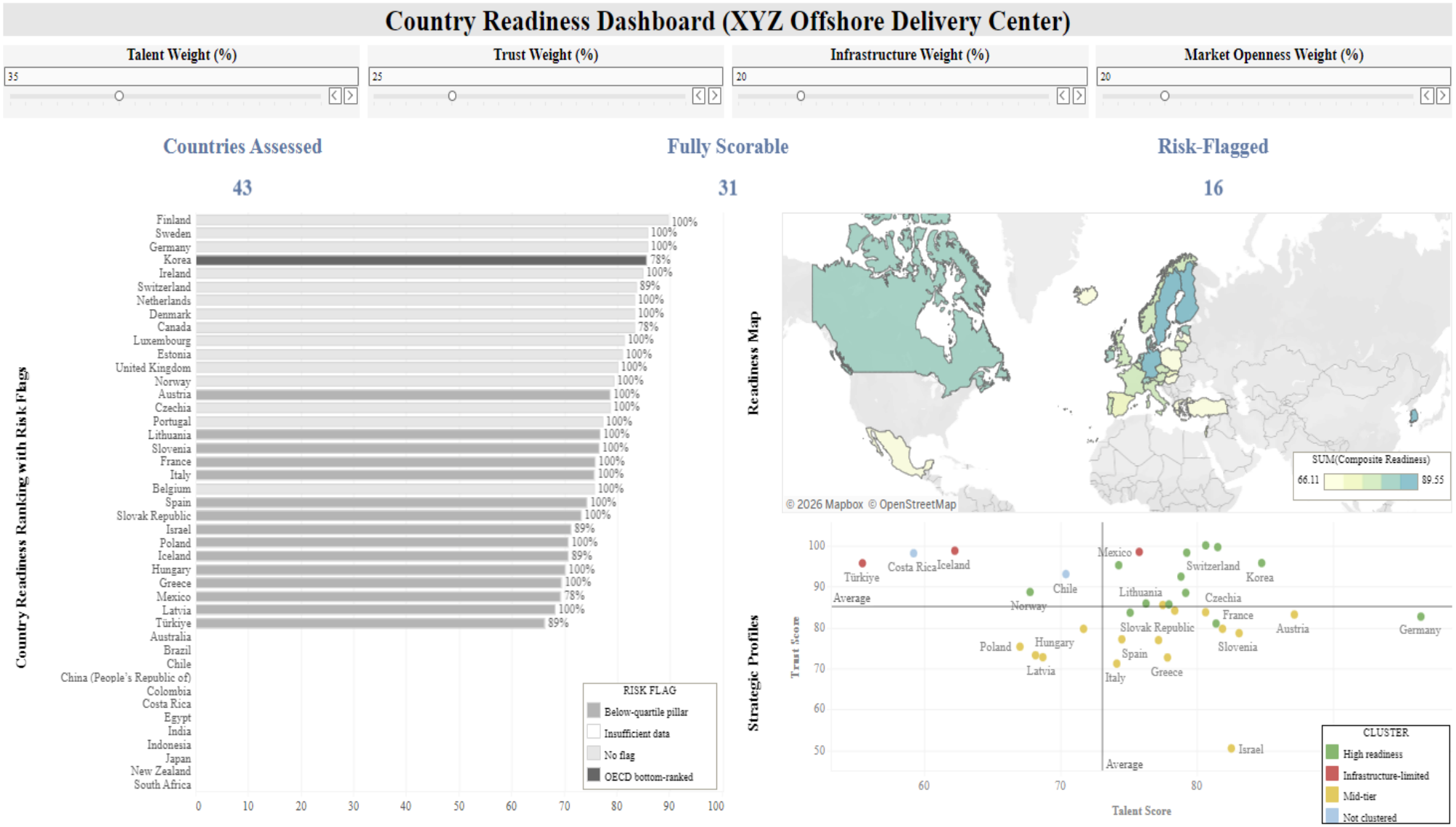


Figure 2. Executive country readiness dashboard. Readiness map, strategic profiles scatter, and composite readiness ranking with risk flags, with adjustable pillar weight controls and summary measures.

The high-readiness cluster comprises Canada, Denmark, Estonia, Finland, Germany, Ireland, Korea, Luxembourg, the Netherlands, Norway, Sweden, Switzerland, and the United Kingdom. Country-level pillar scores for the full candidate set are listed at Figure C7. The infrastructure-limited cluster, comprising Iceland, Mexico, and Türkiye, records a mean infrastructure score of 38.9 against 82.7 for the leaders. For an operation that depends on cloud-hosted EDC platforms, that gap represents a material operational concern and would require substantial compensating investment, despite those countries' strong trust scores.

Sensitivity Testing: Because the pillar weights represent management judgment rather than measured fact, the ranking was recalculated under four weighting scenarios.

Scenario	Weights (Talent / Trust / Infrastructure / Market Openness)	Top five
Baseline	35 / 25 / 20 / 20	Finland, Sweden, Germany, Korea, Ireland
Security-first	25 / 40 / 15 / 20	Finland, Sweden, Korea, Switzerland, Canada
Equal weights	25 / 25 / 25 / 25	Finland, Sweden, Ireland, Korea, Netherlands
Talent-heavy	50 / 20 / 15 / 15	Germany, Finland, Korea, Sweden, Ireland

Table 3. Top five countries under four weighting scenarios.

Finland, Sweden, and Korea appear in the top five under all four scenarios; Ireland under three. Germany moves from third under baseline weights to first when talent is prioritized (Figure C2), but falls to eighth under security-first weights and seventh under equal weights. It therefore appears in the top five under only two of the four scenarios, making its position markedly more sensitive to management priorities than those of Finland, Sweden, or Korea. Finland holds first place in three of four scenarios and second in the fourth.

This test separates countries that are genuinely strong from countries that happen to suit one set of assumptions, and it is the basis on which the shortlist below is constructed.

Risk Screening: Sixteen of the 31 scorable countries carry at least one risk signal. Fifteen fall within the bottom quartile on Trust, Market Openness, or Infrastructure; Greece and Latvia are flagged on all three. One country, Korea, is classified by the OECD among the lowest-ranked on a core indicator.

Two observations follow. First, none of the 13 high-readiness countries carries a below-quartile flag, which means the clustering and the risk screen, constructed independently, agree. Second, Korea's flag is specific and consequential: its digitally-deliverable services exports score 30.95, against 85.7 for Ireland and 60.2 for Finland. For a center whose entire output crosses borders digitally, this is the single indicator in the dataset most directly tied to the operating model, and it is the one on which Korea performs worst.

Evidence Gaps: Twelve of the 43 candidate countries could not be scored because they lack data for an entire pillar. These are Australia, Brazil, Chile, China, Colombia, Costa Rica, Egypt, India, Indonesia, Japan, New Zealand, and South Africa. Their data completeness ranges from 33 percent (Egypt) to 78 percent (Brazil, Colombia).

The composition of this group matters more than its size. It contains most of the emerging markets a clinical data services firm would ordinarily evaluate. India, for instance, records a Talent score of 70.9 and Market Openness of 64.5, both competitive with mid-tier European countries, but has no Trust or Infrastructure data at all. The absence of a score is a statement about OECD coverage, not about India.

The scorable countries are also geographically concentrated, as Figure C5 shows: almost the entire ranked field sits in Europe, with Canada, Korea, Mexico, Israel, and Türkiye the only exceptions. The practical consequence is that this analysis identifies the best location among well-documented economies. It cannot, on its own, establish that those economies are the best locations overall.

Supplementary Factors Beyond the OECD Dataset: Two factors outside the OECD dataset were assembled for the shortlisted, conditional, and representative data-sparse countries. English-language proficiency is taken from the EF English Proficiency Index 2025 (EF Education First, n.d.), measured on a 0–800 scale; Ireland is not ranked because the index covers only countries where English is not a primary language. Time-zone overlap is calculated from standard UTC offsets as the number of hours a local 09:00–17:00 working day coincides with 09:00–17:00 U.S. Eastern Time. Comparable labor-cost data was not available at the time of analysis and is identified in Recommendation 3 for acquisition; no estimate is presented here, because an unverified cost figure would carry more weight in a location decision than its reliability justifies.

Country	EF English 2025 (0–800)	Overlap with U.S. Eastern (hrs)	Status
Finland	590	1	Shortlist
Sweden	608	2	Shortlist
Ireland	Native (not ranked)	3	Shortlist
Korea	523	0	Conditional
Estonia	578	1	Conditional
Germany	598	2	Reference
Poland	588	2	Mid-tier
Portugal	605	3	Mid-tier
India	490	0	Data-sparse
Brazil	466	6	Data-sparse
Colombia	485	8	Data-sparse
Costa Rica	534	7	Data-sparse

Table 4. Supplementary factors for shortlisted, conditional, and representative data-sparse countries. Time-zone overlap uses standard UTC offsets and varies with daylight saving.

Two points follow. Korea has no overlap with U.S. Eastern business hours, which suits a follow-the-sun handover model but not real-time collaboration with U.S. project managers, a second consideration alongside its risk flag. And the Latin American data-sparse markets record six to eight hours of overlap against one to three for every European candidate, with English proficiency comparable to Korea's. Neither pattern is visible in the OECD readiness data, and both bear directly on the operating model.

Recommendations

Recommendation 1: Advance Finland, Sweden, and Ireland to full due diligence. These three countries combine high composite scores (89.5, 85.7, and 84.8), complete data across all nine indicators, membership of the high-readiness cluster, and no risk flags (Figures 1 and 2). Finland and Sweden remain in the top five under every weighting tested, and Ireland under three of four, so the shortlist does not depend on the specific weights chosen. Ireland additionally records the strongest Market Openness score of the three (92.2), which is directly relevant to establishing a foreign-owned entity and delivering services across borders. Due diligence should address labor cost, English-language proficiency, and the depth of the local clinical-research workforce, none of which this dataset measures.

Recommendation 2: Retain Korea and Estonia as conditional candidates, pending targeted validation. Korea ranks fourth and is robust across all four weighting scenarios (Table 3), but carries a specific and material weakness, shown as the OECD bottom-ranked bar in Figure 2: an OECD bottom-ranked position on digitally-deliverable services exports, scoring 30.95 where the shortlisted countries score between 60 and 92. Its data completeness is also 78 percent rather than 100. Korea should not be eliminated on this

basis, but it should not enter full due diligence until the regulatory and commercial reasons for that score are understood, since cross-border digital service delivery is the center's core function.

Estonia warrants conditional advancement for the opposite reason. It ranks eleventh (80.9) with complete data, no risk flags, and the most evenly balanced pillar profile of any high-readiness country, with every pillar between 80.3 and 81.4. Its position within the high-readiness cluster is shown at Figure C3. It records an EF English score of 578, comparable to the shortlisted countries, and a single hour of overlap with U.S. Eastern business hours, the same as Finland. As a smaller and more recently developed economy within the high-readiness cluster, Estonia is the most plausible candidate for combining strong readiness with the cost efficiency that motivates the expansion. That proposition, however, rests on an expectation about relative wage levels rather than on evidence in this analysis. Advancement should therefore be conditional on obtaining verified labor-cost data for Estonia alongside the shortlisted countries.

Recommendation 3: Commission supplementary data before excluding any data-sparse market. Twelve countries, including India, Brazil, Colombia, and Costa Rica, were excluded from ranking by absence of evidence rather than by weak performance (Figures C6 and C8). Because cost efficiency is a primary driver of the expansion, and labor-cost data sits outside the OECD framework entirely, excluding these markets on the current evidence risks eliminating potentially competitive locations before their cost position has been established. That would be a decision made by default rather than by analysis.

The supplementary factors in Table 4 reinforce the point. Colombia and Costa Rica record eight and seven hours of overlap with U.S. Eastern business hours respectively, against one to three for the European shortlist, with English proficiency scores close to Korea's. Neither can be ranked on readiness, but neither should be dismissed on that basis.

Four markets should be prioritized on the evidence that does exist. Costa Rica records the strongest available profile of the group, with a Trust score of 98.1, Market Openness of 78.4, and seven hours of overlap with U.S. business hours. Colombia offers the widest overlap of any candidate at eight hours, with Market Openness of 73.0. India records the highest Talent score among the data-sparse markets at 70.9, alongside Market Openness of 64.5. Brazil combines an Infrastructure score of 83.6 with 78 percent data completeness, the joint-highest of the group. For these four, management should commission verified labor-cost data, governance-stability measures (World Bank, n.d.), national data-protection assessments (UNCTAD, n.d.), and clinical-research-activity counts, then re-run the readiness comparison on the enlarged evidence base before the shortlist is finalized. The dashboard supports this directly: the framework and weights remain unchanged, and only the underlying data is extended.

5. Discussion / Conclusion

Why a Data-Based Approach Was Essential: The analysis repeatedly produced conclusions that intuition would have missed. Second through ninth place are separated by fewer than three points, so any confident claim that one of them is clearly best would not survive examination. Korea, which a composite score alone would place fourth and recommend, carries a weakness on precisely the dimension that matters most for cross-border delivery. The duplicate-record correction, a purely technical fix, moved a country two places in the ranking. And an early version of the risk measure would have reported 25 at-risk countries rather than 16, because it confused missing data with poor performance.

None of these would have been visible in a judgment-based assessment. Each was surfaced by making the reasoning explicit enough to be checked.

Strategic Impact: For XYZ, the practical value is that the field of 43 countries has been reduced to a defensible shortlist of three, with two conditional candidates and a clearly specified set of further inquiries. Due-diligence spending can now be concentrated where it is most likely to pay off, and leadership can see not only which countries rank highly but why, and where each one is weak.

The analysis also changes what the organization knows about its own decision. The concentration of scorable countries in high-income Europe means that the readiness evidence and the cost rationale for the expansion point in different directions. That tension is not resolved by this dataset, and naming it is more useful to management than a shortlist that conceals it.

A Framework for Long-Term Decisions: The framework developed here is reusable. The pillars are defined by operating requirements rather than by this dataset, the weights are adjustable rather than embedded, and data completeness is reported alongside every score. If XYZ later evaluates a second site, revisits this decision as OECD data is updated, or applies different strategic priorities, the same structure applies without rebuilding.

Its most transferable feature is the separation of risk from evidence gaps. A country that performs poorly and a country about which little is known require different responses: one is a reason for caution, the other a reason to gather more information. Conflating them, as the first version of the risk measure did, systematically penalizes exactly the emerging markets an expansion of this kind is most likely to need.

Next Steps: Four actions are recommended. Commission full due diligence on Finland, Sweden, and Ireland, covering labor cost, English-language proficiency, clinical-research workforce depth, data-protection regime, and entity-establishment requirements. Commission a targeted review of Korea's cross-border digital services position, and of Estonian labor costs, to determine whether either conditional candidate should be elevated. Acquire supplementary data for Costa Rica, Colombia, India, and Brazil, and re-run the comparison on the enlarged evidence base. And set a review point at which the shortlist is confirmed against the enlarged evidence base, before any capital commitment is made.

On the present evidence, Finland, Sweden, and Ireland represent the strongest available options for XYZ's first offshore clinical data delivery center. That conclusion is offered as the best supported by the data currently available, and the framework is designed to be revisited as that evidence improves.

References

EF Education First. (n.d.). EF English Proficiency Index. Retrieved October 8, 2026, from https://www.ef.com/epi

International Labour Organization. (n.d.). ILOSTAT. Retrieved October 8, 2026, from https://ilostat.ilo.org

National Library of Medicine. (n.d.). ClinicalTrials.gov. Retrieved October 8, 2026, from https://clinicaltrials.gov

OECD. (2025). Going Digital Toolkit [Data set]. Retrieved October 8, 2026, from https://goingdigital.oecd.org

UNCTAD. (n.d.). Data protection and privacy legislation worldwide. Retrieved October 8, 2026, from https://unctad.org/page/data-protection-and-privacy-legislation-worldwide

World Bank. (n.d.). Worldwide Governance Indicators. Retrieved October 8, 2026, from https://www.worldbank.org/en/publication/worldwide-governance-indicators

Appendix A. Core Indicators by Strategic Pillar

Coverage is the number of candidate countries (of 43) with a Distance score for each indicator.

Pillar	Indicator (OECD)	Policy dimension	Coverage	Relevance to XYZ
Talent and Workforce	New tertiary graduates in science, technology, engineering and mathematics as a share of new graduates	Jobs	42	Future pipeline of analytical and technical staff
	Digital-intensive sectors' share in total employment	Jobs	42	Workforce already experienced in digital work
Trust and Data Security	Share of enterprises in which own employees carry out ICT security related activities	Trust	25	In-house cybersecurity maturity
	Share of Internet users experiencing abuse of personal information or privacy violations	Trust	32	Prevalence of privacy risks experienced by users
Business Digital Infrastructure	Share of businesses with broadband contracted speed of 30 Mbps or more	Access	30	Reliable access to cloud-based data platforms
	Share of businesses purchasing cloud services	Use	36	Business familiarity with cloud workflows
Market Openness	OECD Foreign Direct Investment Regulatory Restrictiveness Index	Market openness	43	Ease of establishing a foreign-owned entity
	OECD Digital Services Trade Restrictiveness Index	Market openness	42	Ease of delivering digital services across borders
	Digitally-deliverable services as a share of commercial services trade	Market openness	43	Existing digital services export capability

Supplementary indicators examined but excluded from core scoring on coverage grounds: ICT task-intensive jobs (29), employment-based training (28), health data sharing intensity (21).

Appendix B. Data Preparation and Scoring Framework

Component	Approach applied	Purpose
Candidate countries	Excluded United States (home market) and aggregate rows (EU27, G20, OECD average)	Restrict ranking to 43 candidate countries
Measure type	Filtered to Type = "Distance"	Common 0–100 scale across indicators
Score direction	Verified: higher Distance = more favorable, including inverse indicators	Consistent interpretation
Duplicate records	Deduplication flag applied as a context filter	Prevent double-counting (42 affected pairs)
Pillar assignment	Nine core indicators grouped into four pillars	Structure the analytical model
Pillar score	Average of available core indicators within each pillar	Summarize pillar performance
Composite Readiness	Weighted combination of pillar scores via Tableau parameters, normalized by weight sum	Overall comparison and sensitivity testing
Fully scorable	At least one core indicator in every pillar (31 countries)	Eligibility for a composite score
Data-sparse	No core indicators in one or more pillars (12 countries)	Retained for separate assessment
Data completeness	Share of the nine core indicators available per country	Show strength of underlying evidence
Risk flag	OECD Bottom 1–3 on core Trust or Market Openness indicators; or bottom quartile on Trust, Market Openness or Infrastructure	Identify weak observed performance
Insufficient data	Applied where no composite score can be calculated	Separate evidence gaps from risk
Clustering	k-means on four pillar scores, 31 scorable countries	Identify strategic profiles

Appendix C. Supporting Tableau Views

All figures are the author's Tableau analysis of the OECD Going Digital Toolkit dataset (OECD, 2025). The composite readiness ranking and the executive dashboard appear as Figures 1 and 2 in the Results section.

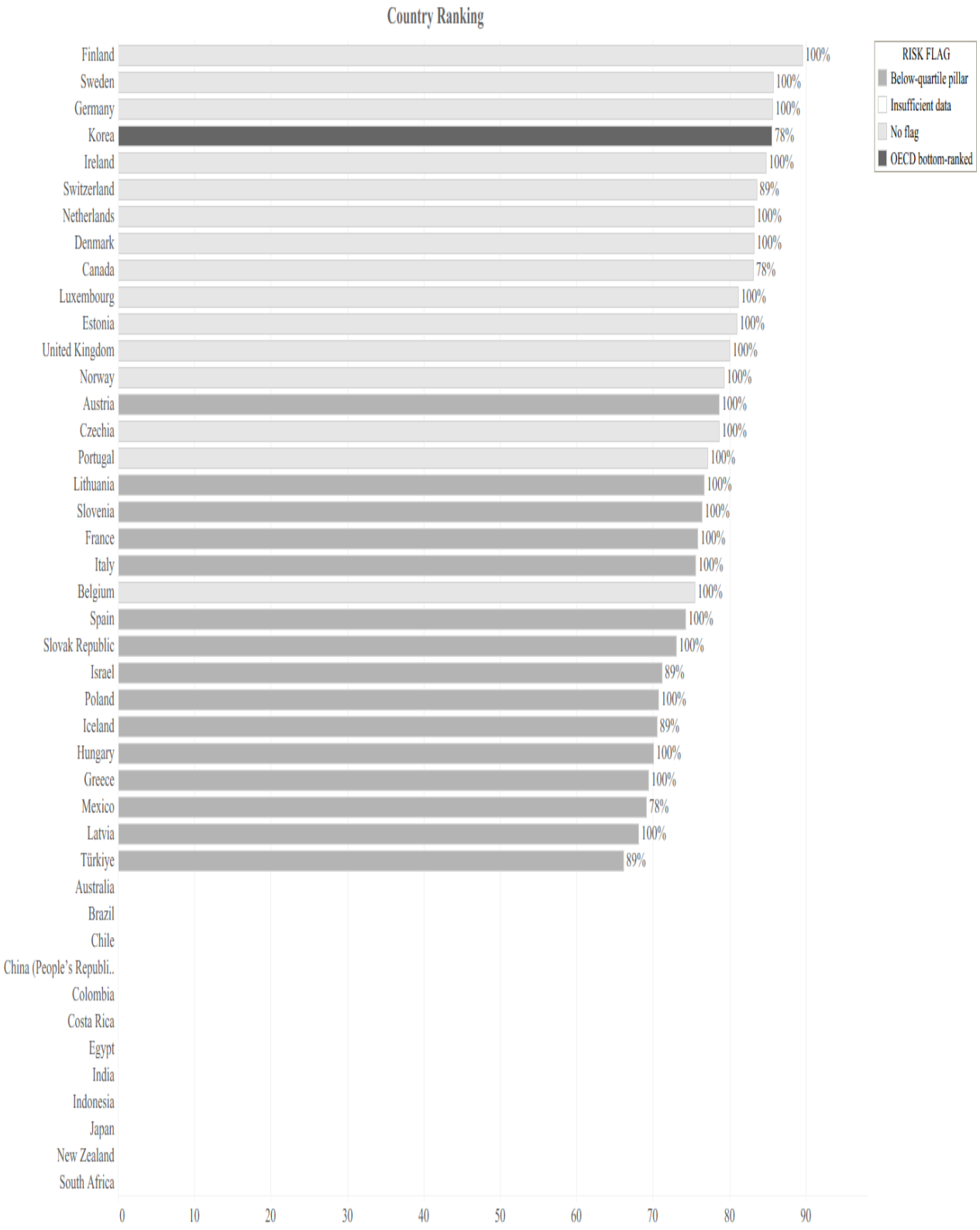


Figure C1. Country readiness ranking under baseline weights. All 43 candidate countries; bar length is composite readiness, color is risk classification, and labels show data completeness. The twelve countries without bars lack a composite score.

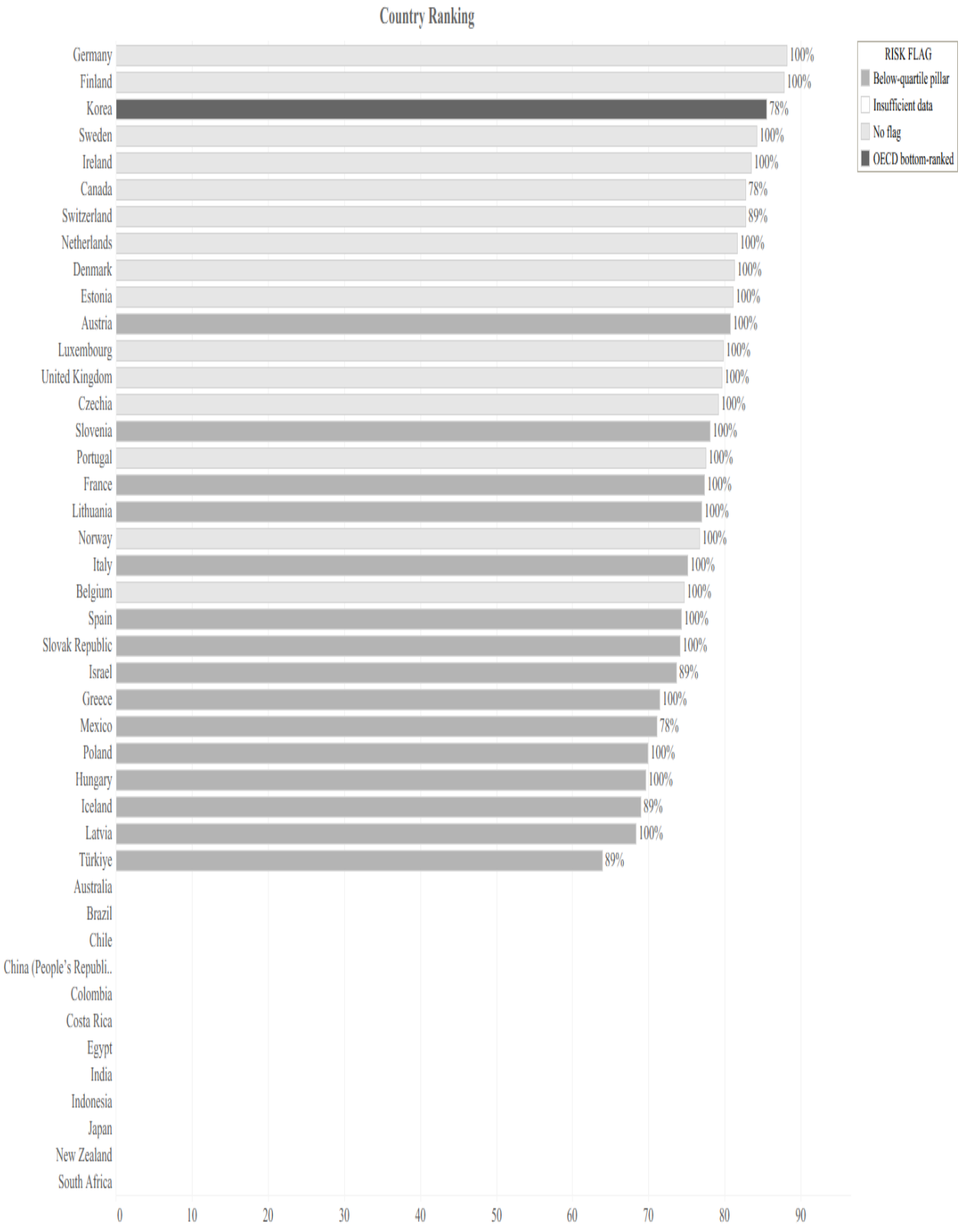


Figure C2. Country ranking under talent-heavy weights. Composite readiness at Talent 50%, Trust 20%, Infrastructure 15%, Market Openness 15%, illustrating the sensitivity of Germany's position.

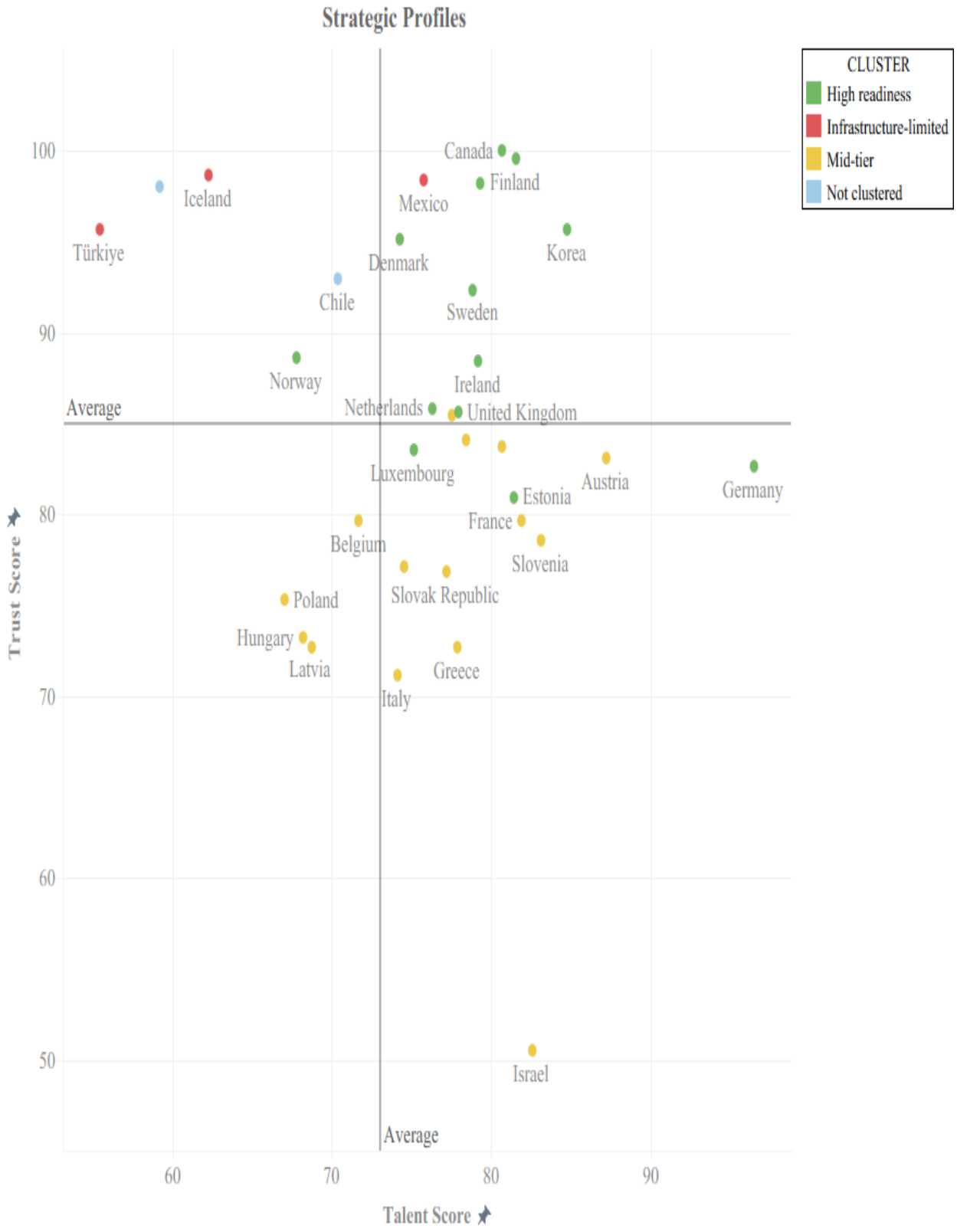


Figure C3. Strategic profiles. k-means clusters plotted on Talent and Trust pillar scores, with quadrant reference lines at the mean of each axis.

Cluster Membership

Cluster Profile	Country	Avg. Infrastructure Score	Avg. Market Openness Score	Avg. Talent Score	Avg. Trust Score
High readiness	Canada	69.23	80.24	80.67	100.00
	Switzerland	70.99	85.35	79.28	98.22
	United Kingdom	71.25	85.31	77.97	85.61
	Luxembourg	71.55	98.33	75.12	83.59
	Germany	76.16	79.80	96.47	82.63
	Estonia	80.62	80.34	81.43	80.95
	Ireland	82.54	92.22	79.20	88.42
	Korea	88.77	70.98	84.78	95.69
	Norway	90.08	76.91	67.78	88.62
	Netherlands	91.50	83.90	76.29	85.82
	Sweden	91.52	83.55	78.86	92.37
	Denmark	94.37	72.74	74.28	95.16
	Finland	96.81	83.75	81.55	99.57
Infrastructure-limit..	Mexico	17.12	73.03	75.79	98.42
	Türkiye	47.70	66.22	55.47	95.65
	Iceland	51.84	68.60	62.25	98.69
Mid-tier	Greece	53.00	66.96	77.87	72.68
	Latvia	58.76	70.66	68.72	72.74
	Slovak Republic	59.36	74.66	77.21	76.89
	France	61.53	74.62	81.91	79.65
	Hungary	66.63	72.73	68.18	73.22
	Austria	66.63	70.25	87.18	83.13
	Slovenia	68.64	69.71	83.12	78.59
	Spain	70.42	73.91	74.51	77.11
	Lithuania	70.95	69.72	77.52	85.48
	Portugal	71.30	71.88	78.40	84.08
	Czechia	72.14	75.17	80.66	83.70
	Israel	74.52	73.70	82.55	50.52
	Poland	75.00	66.88	67.04	75.29
	Belgium	78.17	74.32	71.70	79.65
	Italy	83.23	75.63	74.14	71.18
Not clustered	Chile		69.82	70.41	92.98
	China (People's Republ..		59.81	68.99	
	Costa Rica		78.35	59.23	98.07
	Egypt		49.44	30.56	
	India		64.49	70.92	
	Indonesia		58.70	57.43	
	South Africa		69.28	62.16	
	Japan	30.72	83.59	71.13	
	Colombia	55.36	72.99	67.89	
	New Zealand	71.95	73.39	73.87	
	Brazil	83.63	70.95	62.61	
	Australia	92.48	70.62	69.04	

Figure C4. Cluster membership and mean pillar scores. Segmentation of the 31 fully scorable countries into three strategic profiles, with the 12 data-sparse countries shown as not clustered.

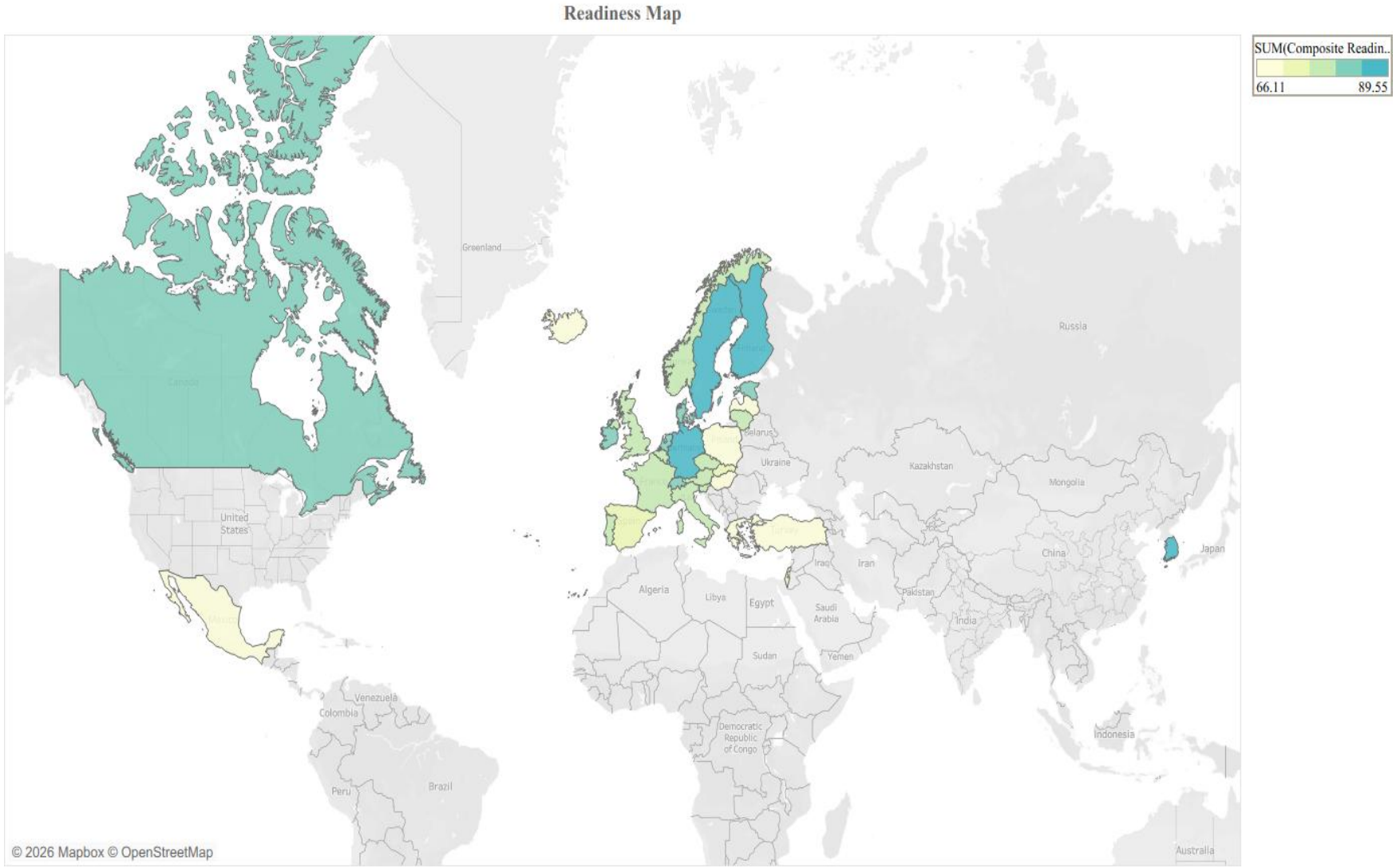


Figure C5. Composite readiness map. Geographic distribution of composite readiness across the 31 fully scorable countries. Unshaded countries lack sufficient data to be scored.

Indicator Heat Map

Country	Infrastructure		Market Openness			Talent		Trust	
	Broadband ≥30 Mbps	*Cloud adoption*	*Digital services exports*	*Digital trade openness*	*FDI openness*	*Digital employment*	*STEM graduates*	*ICT security*	*Privacy violations*
Finland	93.62	100.00	60.24	92.00	99.00	85.10	78.00	100.00	99.15
Sweden	91.56	91.48	61.66	90.00	99.00	82.32	75.40	87.29	97.46
Germany	92.29	60.03	48.41	92.00	99.00	92.93	100.00	66.96	98.30
Korea		88.77	30.95	92.00	90.00	85.30	84.26		95.69
Ireland	84.48	80.60	85.65	92.00	99.00	84.68	73.72	77.45	99.40
Switzerland	84.35	57.62	69.04	98.00	89.00	88.29	70.27		98.22
Netherlands	92.28	90.71	58.71	94.00	99.00	100.00	52.58	73.98	97.66
Denmark	100.00	88.75	25.22	94.00	99.00	84.34	64.23	92.75	97.57
Canada		69.23	55.73	100.00	85.00	89.58	71.76		100.00
Luxembourg	95.79	47.31	100.00	96.00	99.00	96.53	53.72	70.77	96.41
Estonia	86.43	74.81	46.03	96.00	99.00	86.16	76.70	63.57	98.32
United Kingdom	66.62	75.88	57.92	98.00	100.00	92.42	63.52	73.89	97.34
Norway	89.03	91.12	33.73	98.00	99.00	76.30	59.25	79.76	97.48
Austria	73.89	59.37	32.75	84.00	94.00	89.00	85.36	68.36	97.90
Czechia	84.06	60.22	38.51	88.00	99.00	89.09	72.24	67.41	100.00
Portugal	94.71	47.90	26.64	90.00	99.00	79.13	77.67	71.40	96.76
Lithuania	92.87	49.04	17.15	94.00	98.00	82.39	72.66	71.40	99.56
Slovenia	85.91	51.36	27.12	86.00	96.00	86.28	79.96	57.81	99.36
France	88.88	34.18	35.87	90.00	98.00	91.54	72.27	61.51	97.79
Italy	88.05	78.41	42.89	91.00	93.00	84.98	63.31	47.52	94.83
Belgium	90.32	66.02	35.95	88.00	99.00	94.32	49.08	61.51	97.78
Spain	95.08	45.77	40.72	82.00	99.00	90.94	58.09	58.45	95.77
Slovak Republic	74.76	43.96	31.97	94.00	98.00	92.49	61.93	55.58	98.21
Israel	84.10	64.95	46.09	86.00	89.00	89.95	75.16	50.52	
Poland	78.90	71.11	32.64	74.00	94.00	79.78	54.30	52.60	97.99
Iceland	37.98	65.71	43.79	77.00	85.00	73.78	50.72		98.69
Hungary	71.52	61.74	33.19	87.00	98.00	93.09	43.27	49.05	97.39
Greece	75.88	30.13	16.89	86.00	98.00	79.18	76.55	46.21	99.14
Mexico		17.12	42.10	92.00	85.00	79.50	72.09		98.42
Latvia	71.85	45.68	30.99	82.00	99.00	83.46	53.97	46.00	99.48
Türkiye	74.47	20.93	34.67	75.00	89.00	68.48	42.47		95.65
Australia		92.48	32.86	98.00	81.00	80.71	57.37		
Brazil	79.54	87.73	32.86	86.00	94.00	76.38	48.85		
Chile			38.46	74.00	97.00	81.05	59.77		92.98
China (People's Republ..			37.43	65.00	77.00	68.99			
Colombia	27.53	83.19	49.96	72.00	97.00	70.12	65.66		
Costa Rica			43.05	96.00	96.00	76.25	42.21		98.07
Egypt			22.87		76.00		30.56		
India			43.47	72.00	78.00	46.91	94.92		
Indonesia			35.10	69.00	72.00	60.65	54.21		
Japan		30.72	59.76	96.00	95.00	87.87	54.40		
New Zealand		71.95	44.17	90.00	86.00	81.77	65.97		
South Africa			40.83	70.00	97.00	75.71	48.61		

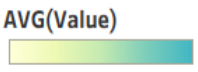


Figure C6. Indicator heat map. Distance scores for the nine core indicators across all 43 candidate countries, ordered by composite readiness. Blank cells identify evidence gaps.

Pillars Score

Country	*Trust Score*	*Talent Score*	*Infrastructure Score*	*Market Openness Score*
Australia		69.04	92.48	70.62
Austria	83.13	87.18	66.63	70.25
Belgium	79.65	71.70	78.17	74.32
Brazil		62.61	83.63	70.95
Canada	100.00	80.67	69.23	80.24
Chile	92.98	70.41		69.82
China (People's Republic of)		68.99		59.81
Colombia		67.89	55.36	72.99
Costa Rica	98.07	59.23		78.35
Czechia	83.70	80.66	72.14	75.17
Denmark	95.16	74.28	94.37	72.74
Egypt		30.56		49.44
Estonia	80.95	81.43	80.62	80.34
Finland	99.57	81.55	96.81	83.75
France	79.65	81.91	61.53	74.62
Germany	82.63	96.47	76.16	79.80
Greece	72.68	77.87	53.00	66.96
Hungary	73.22	68.18	66.63	72.73
Iceland	98.69	62.25	51.84	68.60
India		70.92		64.49
Indonesia		57.43		58.70
Ireland	88.42	79.20	82.54	92.22
Israel	50.52	82.55	74.52	73.70
Italy	71.18	74.14	83.23	75.63
Japan		71.13	30.72	83.59
Korea	95.69	84.78	88.77	70.98
Latvia	72.74	68.72	58.76	70.66
Lithuania	85.48	77.52	70.95	69.72
Luxembourg	83.59	75.12	71.55	98.33
Mexico	98.42	75.79	17.12	73.03
Netherlands	85.82	76.29	91.50	83.90
New Zealand		73.87	71.95	73.39
Norway	88.62	67.78	90.08	76.91
Poland	75.29	67.04	75.00	66.88
Portugal	84.08	78.40	71.30	71.88
Slovak Republic	76.89	77.21	59.36	74.66
Slovenia	78.59	83.12	68.64	69.71
South Africa		62.16		69.28
Spain	77.11	74.51	70.42	73.91
Sweden	92.37	78.86	91.52	83.55
Switzerland	98.22	79.28	70.99	85.35
Türkiye	95.65	55.47	47.70	66.22
United Kingdom	85.61	77.97	71.25	85.31

Figure C7. Pillar scores by country. Talent, Trust, Infrastructure, and Market Openness scores for all 43 candidate countries. Blank cells indicate pillars with no available indicator data.

Data-Sparse Candidates

Country	Avg. Infrastructure Score	Avg. Market Openness Score	Avg. Talent Score	Avg. Trust Score	Avg. Data Completeness %
Australia	92.48	70.62	69.04		67%
Brazil	83.63	70.95	62.61		78%
Chile		69.82	70.41	92.98	67%
China (People's Republ..		59.81	68.99		44%
Colombia	55.36	72.99	67.89		78%
Costa Rica		78.35	59.23	98.07	67%
Egypt		49.44	30.56		33%
India		64.49	70.92		56%
Indonesia		58.70	57.43		56%
Japan	30.72	83.59	71.13		67%
New Zealand	71.95	73.39	73.87		67%
South Africa		69.28	62.16		56%

Figure C8. Data-sparse candidates. The twelve countries excluded from composite scoring, showing the pillar scores that are available and overall data completeness.